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Product having a tab with an embedded memory portion

Abstract

Provided is, among other things, a shoe having a sole that includes multiple layers and an upper attached to and extending above the sole. In addition, the shoe includes a tab having: (1) a first portion, located between adjacent layers of the sole and thus hidden from view, that is imprinted with identification information pertaining to the shoe and (2) a second portion that is visible during ordinary use of the shoe.

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Background/Summary

(1) This application claims the benefit of U.S. Provisional Patent Application Ser. No. 61/108,488, filed on Oct. 25, 2008, and titled “Securely Printed Shoe Tab”, which application is incorporated by reference herein as though set forth herein in full.

FIELD OF THE INVENTION

(1) The present invention pertains, among other things, to footwear (i.e., shoes) that include a securely printed tab, e.g., so as to prevent at least some of the printed information from becoming obscured, either through wear or through intentional destruction, as well as to the tab itself and methods for manufacturing shoes that include such a tab.

BACKGROUND

(2) It sometimes is the case that a manufacturer of shoes wants or is required to include certain information on its products. For example, Section 103 of the Consumer Products Safety Act of 2008, HR 4040, enacted as Public Law No. 110-314, mandates certain labeling of children's products. However, each of the conventional approaches to product labeling has its own drawbacks and shortcomings.

SUMMARY OF THE INVENTION

(3) The present invention addresses this problem by providing, among other things, a shoe that includes a tab having two portions—a printed portion that is embedded in part of the shoe and an exposed portion.

(4) Thus, one embodiment of the invention is directed to a sole that includes multiple layers and an upper attached to and extending above the sole. In addition, the shoe includes a tab having: (1) a first portion, located between adjacent layers of the sole and thus hidden from view, that is imprinted with identification information pertaining to the shoe and (2) a second portion that is visible during ordinary use of the shoe.

(5) By virtue of the foregoing arrangement, e.g., a shoe can be labeled with information that is protected from wear in ordinary use. Preferably, however, the printed hidden portion of the tab can be retrieved when desired.

(6) The foregoing summary is intended merely to provide a brief description of certain aspects of the invention. A more complete understanding of the invention can be obtained by referring to the claims and the following detailed description of the preferred embodiments in connection with the accompanying figures.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) In the following disclosure, the invention is described with reference to the attached drawings. However, it should be understood that the drawings merely depict certain representative and/or exemplary embodiments and features of the present invention and are not intended to limit the scope of the invention in any manner. The following is a brief description of each of the attached drawings.

(2) FIG. 1 is an exploded lateral cross-sectional conceptual view of a representative conventional shoe.

(3) FIG. 2 illustrates an exploded lateral cross-sectional conceptual view of an exemplary shoe having a securely printed tab.

(4) FIG. 3 illustrates an exploded lateral cross-sectional conceptual view of an alternate embodiment of a shoe having a securely printed tab.

(5) FIG. 4 is a cross-sectional view of a portion of a shoe having a securely printed tab according to a further embodiment.

(6) FIG. 5 is a longitudinal cross-section of a portion of a shoe's tongue that includes a securely printed tab.

(7) FIG. 6 is perspective view of a shoe illustrating different exemplary positions at which a securely printed tab can be located.

(8) FIG. 7 is a perspective view of a shoe having a plurality of high-density hook-and-loop (e.g., Velcro™) straps, one of which having a securely printed tab.

(9) FIG. 8 is a perspective view of a sandal-style shoe having a plurality of straps, one of which having a securely printed tab.

DESCRIPTION OF THE PREFERRED EMBODIMENT(S)

(10) The following disclosure provides certain examples of a shoe that includes a securely printed tab. However, it should be understood that the present invention can apply to any type of shoe, including a shoe having a fully enclosed upper, a shoe having an upper with some openings, or a shoe that has a predominantly open upper (such as a sandal). Accordingly, except to the extent expressly stated to the contrary, the term “shoe” as used herein is intended to encompass any kind of footwear.

(11) FIG. 1 is an exploded cross-sectional conceptual view of a representative conventional shoe **10** that includes at least some layered construction. For example, the sole of shoe **10** includes an outsole **12** and an insole **14**. A sock layer **17** is disposed on top of the insole **14**, with the result that three layers are provided beneath the wearer's foot (i.e., outsole **12**, insole **14** and sock layer **17**). At the same time, the shoe's upper includes an inner wall **22** and an outer wall **25**, so that the shoe's sidewalls include two different layers.

(12) The various layers can be joined together in any of a variety of different ways, including stitching, gluing, any other kind of bonding, or any combination of any of the foregoing. For instance, outsole **12** and insole **14** can be both glued and stitched together. In addition, as shown in FIG. 1, outer wall **25** might be sandwiched between outsole **12** and insole **14** at its lower end, while inner wall **22** is glued and/or stitched to insole **14** at its lower end, with inner wall **22** and outer wall **25** being stitched together at their upper ends (e.g., using stitching **27**) after folding the top edge **85** of outer wall **25** over the top edge of inner wall **22**. In any event, the precise method for attaching

the various layers that form a particular shoe typically will not be important to the present invention. In addition, as discussed in more detail below, the present invention typically can be used with a wide variety of different kinds of shoes.

(13) Generally speaking, the present invention contemplates the use of a tab that is imprinted with, or otherwise provided with, information identifying (or otherwise pertaining to) the shoe to which the tab is attached. Preferably, the tab is configured as a rectangular piece of thin sheet material, such as nylon, another synthetic material or a natural fabric material. In any event, at least in embodiments in which the exposed part of the tab is located inside the shoe, the tab preferably is relatively soft, such that no, substantially no, or very little discomfort will result when disposed adjacent the wearer's foot. In certain embodiments, the tab has a length of 1-4 cm (centimeters) and a width of 0.5-2.0 cm, although the precise dimensions typically will be determined by the amount of space that is required to accommodate the desired printed indicia. That amount, in turn, typically will depend upon the amount of information to be included, as well as the printing or other encoding method to be used, particularly the size of the characters or other symbols or codes used.

(14) The information that is provided on the tab preferably includes identification information for the shoe to which the tab is attached. More preferably, such information preferably allows identification of the particular manufacturing lot of which the shoe was a part (e.g., by specifying the name of the manufacturer, the location in which the shoe was manufactured, and/or the specific lot number). Accordingly, such information can include a code specifically identifying the manufacturing lot or other information that allows one to determine the manufacturing lot.

(15) Once incorporated into a shoe according to the present invention, the tab preferably includes both a visible portion and a portion that is hidden from view in ordinary use of the shoe. In the preferred embodiments, both such portions include the same or similar identification information. In any event, both such portions preferably include information that permits determination of the manufacturing lot of which the shoe was a part. For example, the portion exposed to view might include a manufacturing lot designation code, e.g., allowing the consumer to easily identify if the shoe has been recalled. At the same time, the information on the portion of the tab that is hidden from view might include a different code that can be interpreted by the manufacturer to identify the manufacturing lot. Also, in alternate embodiments, only the hidden portion of the tab includes the subject (e.g., manufacturing lot) identification information, while the exposed portion includes other information (e.g., the shoe's size and/or the manufacturer's name, trademark or identification information) or is blank.

(16) The ways in which different portions of information are printed or otherwise encoded onto the tab can be the same or can vary depending upon where the information occurs on the tab (e.g., whether it occurs on the hidden portion or the exposed portion of the tab). Either or both can be printed with ink or dye, impressed or surface encoded in any other way. In addition, or instead, either or both can be magnetically encoded (assuming the tab is made of a magnetic material) or encoded using any semiconductor or other memory technology. At the present time, ordinary ink or dye-based printing is preferred, primarily for its cost advantage. However, as future technologies evolve, that preference might change.

(17) FIG. 2 illustrates an exploded cross-sectional conceptual view of an example of a shoe **50** that includes a tab **60** according to the present invention. As shown, the hidden portion **62** of tab **60** is embedded between the insole **14** and the sock layer **17**, such that the visible portion **64** extends upwardly from the sock portion **17**. In certain embodiments, the hidden portion **62** of tab **60** is simply inserted between the insole **14** and the sock layer **17** prior to attaching those layers together (e.g., by gluing or stitching). In other embodiments, tab **60** is first attached to one of such layers (e.g., by gluing) and then the two layers are attached to each other (e.g., by gluing or stitching). Also, it is noted that the hidden portion **62** of tab **60** instead could be sandwiched between insole **14** and outsole **12** or between any other layers in the sole of a shoe.

(18) In one specific embodiment, the desired information is imprinted on the outer side **65** of tab

60, so as to reduce the likelihood that it will be rubbed off during normal wear. Then, the user would only need to fold back tab **60** in order to read it. However, such written indicia instead could be imprinted on the inner side **66** of tab **60**. In either event, it sometimes is preferable to print the desired information on the same side of tab **60** for both the hidden portion **62** and the visible portion **64**, in order to reduce the costs of manufacturing tab **60** (i.e., by using only a single-sided printing).

(19) FIG. 3 illustrates an exploded cross-sectional conceptual view of an alternate embodiment of a shoe **80** includes a tab **90**. In this embodiment, the hidden portion **92** of tab **90** is sandwiched between the inner wall **22** and outer wall **25** of the shoe's upper, so that the visible portion **94** of tab **90** extends downwardly from the top edge **85** of the shoe **80**. In the present embodiment, this is accomplished by simply layering portion **92** of tab **90** between inner wall **22** and outer wall **25** and then applying stitching **27**. Similar to the preceding embodiment, it often is preferable that the written indicia are imprinted on only a single side of tab **90**.

(20) In any event, it is preferable that any side of the tab (e.g., tab **60** or **90**) that is imprinted with information within the hidden portion (e.g., hidden portion **62** or **92**) is not exposed to glue. For example, referring back to FIG. 2, if the written indicia are printed on the outer side **65** of tab **60** (or at least hidden portion **62** of tab **60**), then glue might be applied to the top surface of insole **14**, but not to the bottom surface of sock layer **17**, when bonding those two layers together.

(21) It should be noted that the foregoing examples merely illustrate certain aspects of the invention. In alternate embodiments, the tab that has been imprinted or otherwise encoded with the desired information can be partially sandwiched between any layers, components or portions of the shoe, or even enclosed within a single layer, component or portion of the shoe. In other words, the hidden imprinted portion of the tab can be embedded within any portion of the shoe, although in the preferred embodiments it is permanently embedded within a nonremovable part of the shoe. Also, the portion of the tab so sandwiched, enclosed or otherwise embedded need not be flat, but instead, e.g., can be folded or rolled.

(22) For example, a cross-sectional view of a portion of a shoe according to a further embodiment is shown in FIG. 4, which illustrates a tab **120** having a hidden (imprinted) portion **122** that is rolled up and enclosed within the top portion of a shoe side wall **130**. More specifically, in this embodiment the shoe upper is made of just a single wall **130** (i.e., non-layered), the very top of the edge of which being sewn into a loop using stitching **132**. However, the hidden portion **122** of tab **120** is first rolled up before the side wall **130** is looped around it and stitched. As a result, the exposed portion **124** of tab **120** extends downwardly from the very top edge of side wall **130**.

(23) In the embodiments discussed above, the tab is attached to the bottom of the shoe or some portion of the shoe's upper. In other embodiments, the tab can be attached to any other component(s) or portion(s) of the shoe, such as the tongue (if provided) or a shoe strap (e.g., in the case of a sandal or a shoe having a high-density hook-and-loop, such as Velcro™, tightening strap). As indicated above, the hidden portion (which preferably includes printed or otherwise encoded information) can be sandwiched between layers forming such component(s) or portion(s) or can be enclosed within a single one of such component(s) or portion(s).

(24) Another example of a way in which a securely printed tab can be attached to a shoe according to the present invention is illustrated in FIG. 5. More specifically, FIG. 5 shows a portion of a shoe's tongue **160**, including a top layer **162**, a bottom layer **164** and securely printed tab **165**. As shown, top layer **162** folds over the bottom of bottom layer **164**, the printed hidden portion **167** of tab **165** is inserted between the top layer **162** and the bottom layer **164**, tab **165** is looped back so that hidden portion **167** and exposed portion **168** are parallel and adjacent to each other, and then tongue **160** is sewn up by stitching **170** through top layer **162**, tab **165**, bottom layer **164**, and then the folded-over portion of top layer **162**. Preferably, the interior of tongue **160** (between the top layer **162** and the bottom layer **164**) is filled with padding material.

(25) FIG. 6 illustrates different exemplary positions at which a securely printed tab according to the

present invention can be located on a shoe **200**. For instance, tab **60** (having a hidden imprinted portion that is disposed between layers of the shoe's sole and a visible portion that extends vertically up along the inside surface of the shoe's upper) has been described above in connection with FIG. **2**. Tab **90** having a hidden imprinted portion that is sandwiched between the inner and outer walls of the shoe's upper and a visible portion that extends downwardly from the top edge of the shoe, parallel and adjacent to the hidden portion) has been described above in connection with FIG. **3**. Tab **120** (having a hidden imprinted portion that is rolled up and enclosed within the top edge of the shoe's upper) has been described above in connection with FIG. **4**. Tab **202** is similar to tab **120**, but is attached further down from the top edge of the shoe's upper, so that the hidden portion can extend upwardly in parallel to the shoe's side wall.

(26) Tab **204** is similar to tab **202**, but instead of attaching close to the top edge of the shoe's upper, the visible portion of tab **204** protrudes from a location near the middle of the shoe's side wall, with the hidden portion being sandwiched between layers **207** and **208** of the shoe's side wall. Note that in the present example layer **207** only overlaps a relatively small portion (significantly less than half) of layer **208**. Tab **210** is similar to tab **204**, but the hidden printed portion of tab **210** is sandwiched between two layers that overlap near the front of the shoe **200**, and the visible portion of tab **210** extends upwardly along the shoe's upper. Tab **212** also is similar to tab **204**, but the hidden portion of tab **212** is disposed between the layer forming the front section of the shoe **200** and a layer **216** that is part of the lacing section of the shoe **200**.

(27) Tab **220** also is similar to tab **204**, but the hidden portion of tab **220** is disposed between layers of the shoe's tongue. However, unlike tab **165**, the visible portion of tab **220** does not run along the shoe's tongue **222**, but rather extends straight out and away from it.

(28) Tab **230** attaches to the top rear of the shoe's upper, with the hidden printed portion disposed between layers of the shoe's upper. In addition, the visible portion of tab **230** is formed into a loop in order to assist in pulling the shoe **200** onto the wearer's foot. It is noted that the visible portion of any of the other tabs discussed herein also can be formed into a loop, in which case the hidden printed portion of the tab typically will have two segments, corresponding to the two ends of the loop, and either or both of such segments can be imprinted with information, such as any of the identification information described above. For example, tab **220** can be formed into a loop so as to assist the user in pulling up tongue **222**.

(29) In addition to attaching a securely printed tab according to the present invention to other parts of the shoe, e.g., as indicated above, such a tab can be attached to laces, straps or other components of the shoe that are used for tightening the shoe around the wearer's foot. Preferably, however, such laces, straps or other components are non-removable parts of the shoe.

(30) An example is illustrated in FIG. **7**, which shows a shoe **250** having a plurality of high-density hook-and-loop (e.g., Velcro™) straps **253-255** (e.g., having high-density hook or loop material along the entire underside of each of straps **253-255**). Attached to strap **255** is a securely printed tab **260** that includes an exposed portion **262** and a hidden imprinted portion **263**. In the present embodiment, the hidden imprinted portion **263** is disposed between the top layer (e.g., made a nylon) and the bottom layer (made of high-density hooks or loops) of strap **255**, with the hidden imprinted portion **263** having been placed between such top and bottom layers, and then the top and bottom layers having been stitched together around the periphery of strap **255**.

(31) In a similar embodiment, illustrated in FIG. **8**, a securely printed tab **300** is attached to the strap **302** of a sandal **305**. Strap **302** can employ any kind of tightening mechanism, but often will use mating high-density hook-and-loop materials (e.g., with a pad **307** of such material at the distal end of each strap. In the present embodiment, tab **300** attaches to strap **302** in a manner similar to the way that strap **90** attaches to the upper of shoe **80** (as discussed above in connection with FIG. **3**).

(32) Generally speaking, each of the embodiments described above concerns a shoe that includes a tab having two portions: a printed portion that is hidden from view by being permanently

embedded within some preferably non-removable part of the shoe and a visible portion which also may include some printed information (the same or different than the information printed on the hidden portion). At the same time, in the preferred embodiments, the hidden portion of the tab is embedded within the shoe using a technique, such as stitching (which generally is preferred), or gluing where the adhesive material does not come into contact with the printed information, that permits retrieval of the hidden portion of the tab and reading of the information imprinted on it, when desired, albeit often with some disassembly or even partial destruction of the shoe.

(33) The foregoing embodiments discuss attachment of a printed or otherwise encoded tab to various portions of a shoe. However, it should be noted that the same kind of printed or otherwise encoded tab can be attached to any other article of clothing or apparel, e.g., using any of the attachment techniques described above.

(34) Additional Considerations.

(35) Several different embodiments of the present invention are described above, with each such embodiment described as including certain features. However, it is intended that the features described in connection with the discussion of any single embodiment are not limited to that embodiment but may be included and/or arranged in various combinations in any of the other embodiments as well, as will be understood by those skilled in the art.

(36) Thus, although the present invention has been described in detail with regard to the exemplary embodiments thereof and accompanying drawings, it should be apparent to those skilled in the art that various adaptations and modifications of the present invention may be accomplished without departing from the spirit and the scope of the invention. Accordingly, the invention is not limited to the precise embodiments shown in the drawings and described above. Rather, it is intended that all such variations not departing from the spirit of the invention be considered as within the scope thereof as limited solely by the claims appended hereto.

Claims

1. A product, comprising: a plurality of components; and a tab having a portion, that (i) is permanently embedded within a part of said product that includes at least one of the components, and thereby hidden from view, and (ii) comprises a memory device storing information pertaining to the product.
2. A product according to claim 1, wherein the tab also has a second portion that is visible during ordinary use of the product.
3. A product according to claim 2, wherein the second portion of the tab is configured as a piece of thin sheet material.
4. A product according to claim 3, wherein the second portion of the tab is rectangularly shaped and soft.
5. A product according to claim 2, wherein the second portion of the tab is imprinted with identification information pertaining to the product.
6. A product according to claim 2, wherein the second portion of the tab is formed into a loop.
7. A product according to claim 1, wherein the memory device is encoded with the information at least one of magnetically or in semiconductor form.
8. A product according to claim 1, wherein the portion of the tab is sandwiched between a first component and a second component of the product.
9. A product according to claim 8, wherein each of the first component and the second component is a layer of material.
10. A product according to claim 9, wherein the first component and the second component are attached to each other by at least one of gluing or stitching.
11. A product according to claim 9, wherein the portion of the tab is stitched between the first component and the second component.

12. A product according to claim 9, wherein the tab is attached to at least one of the first component and the second component with glue.
 13. A product according to claim 1, wherein the product includes two adjacent layers attached to each other around their peripheries using stitching, and wherein the portion of the tab is disposed between said two adjacent layers.
 14. A product according to claim 1, wherein the product is a children's product.
 15. A product according to claim 1, wherein at least one of the components comprises a sheet material, and wherein the portion of the tab is enclosed within an edge of the sheet material that has been folded or looped over.
 16. A product according to claim 1, wherein the product is an article of apparel.
 17. A product according to claim 16, wherein the product is an article of footwear.
 18. A product according to claim 1, wherein said part, into which said portion is permanently embedded, is a nonremovable part of said product.
 19. A product according to claim 1, wherein the stored information pertaining to the product identifies a manufacturing lot of which the product was a part.
 20. A product according to claim 1, wherein the stored information pertaining to the product identifies the product.
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